

Institute of Actuaries of India

Subject SP8 – General Insurance: Pricing

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

- i) The fleet has only 2 years of data as opposed to 3 years of data available
- ...and recent year's experience has been unusually heavy on account of possibly one-off event.
 - Thus, reliance on experience based pricing might not be truly accurate to price the add-on for the forthcoming year.
 - The insurer need to use a credibility model
 - ...viz., Classic / Bayesian / Bulmann- Straub Formula and
 - i.e. use both the fleet's own data and the overall portfolio data available with the insurer to arrive at the final premium.
 - The fleet's experience may not be reliable and more weightage has to be given to the insurer's own data.
 - However, the overall portfolio's experience would also be affected for the current year on account of lockdown
 -and the overall portfolio's experience needs to be adjusted.
 - The previous years' experience might be given more weightage and would form the starting point
 - ...with adjustments being made for change in terms and conditions, rate changes etc.

[0.5 each, 5]

- ii) The standard for full credibility is given by:

$$(y/k)^2 * (\sigma_N^2 / \mu_N + \sigma_X^2 / \mu_X^2) \quad [1]$$

$$\Phi(y) = (1+P)/2 = 0.95$$

$$\text{hence, } y = 1.645$$

$$k = 5\%$$

$$\mu_N = 1000$$

$$\sigma_N^2 = 1000 * (1-0.05) = 950 \quad [0.5]$$

$$\mu_X = \exp(8.2198 + 0.5 * 1.4075^2) = 10000 \quad [1]$$

$$\sigma_X^2 = 10000^2 * (\exp(1.4075^2) - 1) = (25000)^2 \quad [1]$$

$$\text{Standard for full credibility} = 7794 \text{ claims} \quad [0.5]$$

[4]

- iii) Standard for Credibility is directly proportional to the Coefficient of Variation for severity. [1]

Coefficient of Variation is defined as (σ_X / μ_X) .

Hence if the losses are capped / large losses are removed, it brings down σ_X more rapidly than μ_X . [1]

With low Coefficient of Variation, the Standard for Credibility reduces. Hence, the lesser number of claims will be required to have more credibility. [1]

[3]**[12 Marks]**

Solution 2:

- The claims data available with the reinsurer for pricing may not be as detailed as that of direct insurer.
- The reinsurer mostly has bordereau data in case of proportional reinsurance
- ...or loss data above a particular limit in case of non-proportional reinsurance
- ...whereas the direct insurer has access to the ground up losses.
- The loss experience of the reinsurance would be generally more volatile as compared to that of the direct insurer.
- Reinsurer may have to consider any changes in inuring reinsurance and its impact on the the claims data.
- Reinsurers tend to use exposure based pricing methodology while pricing non-proportional covers
- e.g. catastrophic models, exposure curves / ILF, etc. rather than Frequency - Severity, Burning cost method, GLM etc.,
- In case of proportional insurance, the reinsurer does not price the risk instead take up the proportion of premium decided by the cedant.
- In this case, the reinsurer concentrates more on Ceding commission and / or Profit commission negotiation with the cedant.
- In direct insurers, except for swing rated products, such situations are rare.
- If there are Limited and / or Paid reinstatements in the reinsurance treaty, the same may have to be priced suitably.
- Due consideration has to be given to whether the claims are indexed and how the inflation rate has been considered. Even though direct insurer also inflate the claims, the effect of indexation will be higher at higher layers.

[0.5 each, **Max 5 Marks**]**Solution 3:**

i)

- It may be analysed if the claims data available with the insurer is divided according to the rating factors viz., whether the data is split according to the location of the property, type of the property, availability of safety systems etc., [0.5]
- It must be confirmed that the data available is complete that it is updated with the latest case outstanding reserve and IBNR. If not there needs to be a loading for the same. Since burglary insurance is a short tailed business, this may not be of major concern in pricing. Moreover, the data is little old. Hence there are more chances that the IBNR claims could have been reported by the time of pricing. [1]
- Number of years of data to be considered in pricing has to be decided. The same depends on the credibility of data and whether the data is relevant to the present conditions. [0.5]
- Any changes in policy conditions viz., change in excess / deductible, change in exclusion, change in underwriting practices, etc., need to be considered. [0.5]
- Claim recording, Claims payment and Claim settlement process have to be checked if they are consistent with the current practices of the insurer. If not, suitable adjustment have to be made [0.5]
- The trend in both frequency and severity has to be studied so that the same may be loaded in the claims data. For example, the propensity to burglary claim might have reduced in a location due to rapid urbanisation. Such trends have to be incorporated in the data. [1]

- Number of years data has to be projected need to be determined as the premiums may have to remain constant due to regulatory / market restrictions for future years. [0.5]
 - Large losses may need to be treated separately from the attrition claims. Large claims definition has to be consistent from year to year. It is essential to remove the large losses / cap the large losses as they may make the analysis volatile. [0.5]
 - If there is need for indexation of large claims so that the same needs to be updated [0.5]
 - There might have been some change in economic, social conditions that could lead to unusually light / heavy experience. Such unusual experience may need adjustment / choose a different base year for pricing. [0.5]
 - Claim inflation rate may be used for inflating the claim to the current level has to be determined. In case of burglary insurance, the sum insured automatically accounts for inflation. Hence premium, if estimated with the exposure measure as sum insured, factors in inflation. However, if the sum insured is not escalated up to date, there may have to be inflation separately. [1]
 - If there is a change in method of distribution, which may impact the claims, the same needs to be incorporated. [0.5]
 - If there are changes in advances in technology, changes in the construction of property – change in material construction, methodology of construction etc., may impact the claims data [0.5]
- [8]**

ii)

- If burning cost method is used, data requirement is lesser. It is enough only if the sum of claims along with Total exposed to risk is available, instead of determining severity, frequency separately. It is simple and quicker than other methods to estimate.
 - If there are no consistent practices with respect to periods of cover, claims management etc., even if the frequency / severity data is available, the same may not be that credible.
 - However, it is quite possible that the insurer has credible data given that the claims data for past 6 years (even if not the most recent) is available. Given that it is a mid-sized insurer it tends to have been established a number of years back, thereby the same may be utilized for analysing the trends in frequency and severity separately.
 - In case of burning cost approach, it may be difficult to adjust past data, as only aggregate data is available. More crude assumptions about reporting delay, settlement delay, inflation rates used for the same may have to be taken, which may decrease the quality of analysis.
- [4]**

iii)

- Commercial property is quite heterogeneous compared to retail households. It is not possible to have similar levels of rating factors for such a cover. [0.5]
- There are several risk factors, such as type of operation carried out in the premises, availability of safety systems, type of objects stored etc., This required detailed inspection by the underwriters and the underwriting judgement plays a major role in rating. [1]
- Given that the past claims may be less in number, credible data may not available for pricing. Given the volatility, it is difficult to establish the technical price. [0.5]
- The properties will have to be individually rated and past claim experience may not be available for pricing. [0.5]

- Brokers and other intermediaries, competitors, impact of insurance cycle have an influence in pricing. [0.5]

[3]

[15 Marks]

Solution 4:

- i) Aliasing occur when there is a linear dependency among the observed co variates $X_1 \dots X_p$. That is one covariate may be identical to some combination of other covariates. [1]

Any suitable example [1]

[2]

ii)

- Though the average annual mileage gives an indication of how long the vehicle has remained on road, it need not be a proxy to determine whether the risk is good / bad as a vehicle may remain only for few minutes per day and might have ended up with accident. [1]
- Also, the driver who is on road for longer time might get more experience and might be less risky. [1]
- It is a factual measure and can be easily measured. [0.5]
- It is not prone to misleading by policyholder unless the distance meter is tampered with. [0.5]
- It does not correlation with many other rating factors viz., age of the vehicle, make and model of vehicle etc., [0.5]
- But the past average annual mileage need not be an indicator of similar annual mileage in the future. The driver might have changed residence / may take lesser trips etc., [0.5]

[Max 3]

- iii) Since the number of pin codes will be quite high in number, it is not advisable to use GLM as there will be so many levels. [1]

In this case Distance based smoothing (a type of Spatial smoothing) is more appropriate.

It incorporates information based on the distance between the location codes. [1]

[2]

iv)

- Lift curves can be used to validate the model
-and is helpful in identifying the model which is more efficient in distinguishing high and low experience.
- First, severity needs to be arrived at for each model point for the models.
- Once the severity is arrived at, since the exposure measure is the same, the same can be placed in the order of increasing severity.
- Severity is plotted against each model point for both the models in the same graph.
- The difference between severity of the model points in the 2 models gives the lift.
- The model which is more steep compared to other is more predictive.

[0.5 each, Max 3]

[10 Marks]

Solution 5:

i)

- Whether the chosen ILF curve is appropriate for the nature of the cover [1]
- the jurisdiction for which the pricing is going to be done. [1]
- Treatment of ALAE in coverage. [0.5]
- Treatment of ULAE and loading. [0.5]
- Whether the curve is for per claimant limit [0.5]
- ...or per occurrence limit. [0.5]
- Effect of trends [0.5]
-and other economic changes [0.5]

[Max 4]

ii)

- In case of property line of business, the upper limit of loss is capped at sum insured.
- The relative loss size distribution can be considered to be independent of the size of the risk.
- In casualty line of business, there is no upper capping on the claims to be paid and hence usually there is no upper limit of the cover.
- Hence it is common to use original loss distribution for deriving ILF curves, instead of relative loss size distribution.

[0.5 each, Max 2]

iii)

- Reibesell curves are used more for Casualty Line of Business. [1]
- Since Casualty line of business claims does not have upper limit, Reibesell curve works around by assuming that each time the sum insured doubles, the loss cost increases by constant factor. [1]

[2]**[8 Marks]****Solution 6:**

i)

- The present formula does not indicate or monitor the penetration i.e. percentage of insured vehicles to total vehicles plying on the road [1]
- The present formula does not ensure increase in penetration under each category of vehicles [1]
- There is higher possibility of less efforts from insurance companies' side to insure 2 wheelers [0.5]
- ...due to low premium per policy as it may not be cost effective [0.5]
- ...and takes higher administrative efforts [0.5]
- Minimum TP obligations will not be known to companies at start of any financial years as it is based on past financial year business [1]
-it takes time to consolidate the industry figures and closing of financial years, thereby making it difficult for the insurer to plan their obligations well in advance [1]

[Max 4]

ii)

- Higher penalties of not having Motor TP insurance [0.5]
- and strict enforcement of law [0.5]
- e.g. no passage through toll points, confiscation of driving license, jail [0.5]

- Purchase of fuel only possible if vehicle is insured for TP [0.5]
- Ensure there is no supply side constraint and its obligatory to write TP business for any insurance company if proposal comes at their door [1]
- Make it mandatory to purchase long term TP insurance cover at time of purchase of new vehicle [1]
- Duration of cover for every vehicle can differ by categories [0.5]
- Minimum TP obligations can be based on number of vehicles in every category and not on premiums volumes [1]
- Minimum TP obligation formula should be such that which achieves 100% insurance over the period and the obligation is known at start of each financial year for every company [1]
- e.g. For year N, it can be based on N-2 business figures than N-1, formula can factor the uninsured vehicle percentage reduction every year [1]

[Max 6]

[10 Marks]

Solution 7:

- Integration of system with other government systems related to motor e.g. ministry of transport, registration authority's system etc. [1]
- Structure or outline of the system [0.5]
- Data structures / formats [0.5]
- E.g. should the policy and claims records be held on separate files [0.5]
- How should the system operate [0.5]
- Who will maintain and can access the system [0.5]
- Specification of technology in conjunction with cost of implementation and maintenance [1]
- e.g. software compatibility with other system [0.5]
- Security of the system - How will the system be monitored? [1]
- Data protection laws should be considered, particularly if the system is to be used by third parties. [1]
- The data that will need to be held. For private motor, we should retain all the data for individual policyholders on the system. [1]
- How data will be uploaded into the system by insurance companies [0.5]
- Age of data i.e. How long the data will be stored. [0.5]
- Statistical analyses will be necessary (in due course) using the data that develops. Every effort should be made to store data indefinitely [0.5]
- Capability of the system: advancement of technology in motor insurance and thus possible future change should be considered [1]
- Ease of use. Need to consider the computer literacy of the end users. [1]
- Implementation of the system. How the system will be developed, and by whom. How it will be tested [1]
- Data reporting requirements [0.5]

[Max 10 Marks]

Solution 8:

- i) The aspects which should be analyzed are:
- Net retention in view of capital position i.e. The amount of risk that an insurer can safely retain, having regard to its solvency position and available capital [1]
 - The extent of the likely exposure to accumulations of risk should be analyzed across the lines and within a line of business. [1]

- Appropriate consideration should be given to aggregate excess of loss cover based on accumulation risk analysis [1]
- The need for catastrophe reinsurance, having regard to the insurer's exposure to both natural and human-made catastrophes [1]
-and the appropriate upper and lower limits for such cover [1]
- The extent of the possible need for reinstatements to cover for different treaties and structures [1]
- Commercial consideration: The value for money provided by the existing reinsurance programme, and whether it meets the objectives of the business [1]
- for example, does it sufficiently reduce the volatility of the claims experience [0.5]
- A general assessment of the appropriateness of existing covers i.e. proportional vs non proportional [1]
- for example, any need of LPT/ADC cover, variable QS arrangements) [0.5]
- The profitability of different layers in non-proportional arrangements [1]
- The effects on capital [0.5]
- The reinsurance cost should be compared with the reduction in capital required and the resultant reduction in cost of capital [1]

[Max 6]

ii)

- Model uncertainty – selection of wrong choice of model.
- Programming error – typing error in model construction and insufficient peer review for programming.
- Incorrect distributional assumptions in modelling claim value uncertainty – distribution assumptions are required to calculate a range of possible values for a claim and wrong assumption is made for claim frequency and severity distribution.
- Other sources of model error – insufficient simulation in a stochastic model, incorrect correlation assumption.

[1 each, **Max 4]****[10 Marks]****Solution 9:**

i)

- Risk factors are any factors that have a bearing on the amount of risk. [1]
- Rating factors used to determine the premium rate for a policy, which is measurable in an objective way and relates to the intensity of the risk [1]

[2]

ii)

- Number of doctors segregated by their specialty is better exposure measure for medical malpractice business [1]
- Business premium volumes alone does not tell about the change in mix of the various specialty doctors [0.5]
- ...which is critical to estimate the frequency [0.5]
- ...and severity of portfolio [0.5]
- Premium volumes alone is incapable of unmasking the rate change in original primary business [1]

[Max 3]**[5 Marks]****Solution 10:**

i)

- virus damage
- hacker attack
- infringement of intellectual property
- slander or libel
- stalking or bullying
- Identity theft
- online identity fraud
- personal data or privacy issues (e.g. loss of or disclosure of private data)
- theft or corporate information or data
- ransomware / malware
- business interruption
- losses relating to breaches or actions by suppliers or business partners

[0.5 each, Max 5]

ii)

- Catastrophe cyber models are at very early stages of development compared to other types of catastrophe models [1]
- Many models will be incomplete focusing on the main types of cyber Risks only [0.5]
- The choice of models available in the market will be limited [1]
- The event module databased will be relatively small due to the limited number of historic cyber events [1]
- ... and the challenges of extrapolating a reliable and comprehensive sets of theoretical events [1]
- ... due to this being a relatively new and developing field of academic research [1]
- Risk itself constantly changes in an arms race [1]
- Correlations between policies hard to asses [1]
- Insurance companies will have very little own data to assist in parameterisation and validation of the model [0.5]
- Difficult to obtain market level data as cyber events are often not publicised by the targets [1]
- Cyber threats are relatively new risk compared to natural catastrophes [1]

[Max 5]

iii)

- type of the industry
- occupation of the insured
- exposure e.g. turnover
- past claims experience
- location of the work force
- frequency of visitors to the site
- the materials handled
- the processes involved
- safety precautions in place
- size of deductibles
- sum insured
- risk management practices and its robustness
- country of operation

[0.5 each, Max 5]**[15 Marks]**
